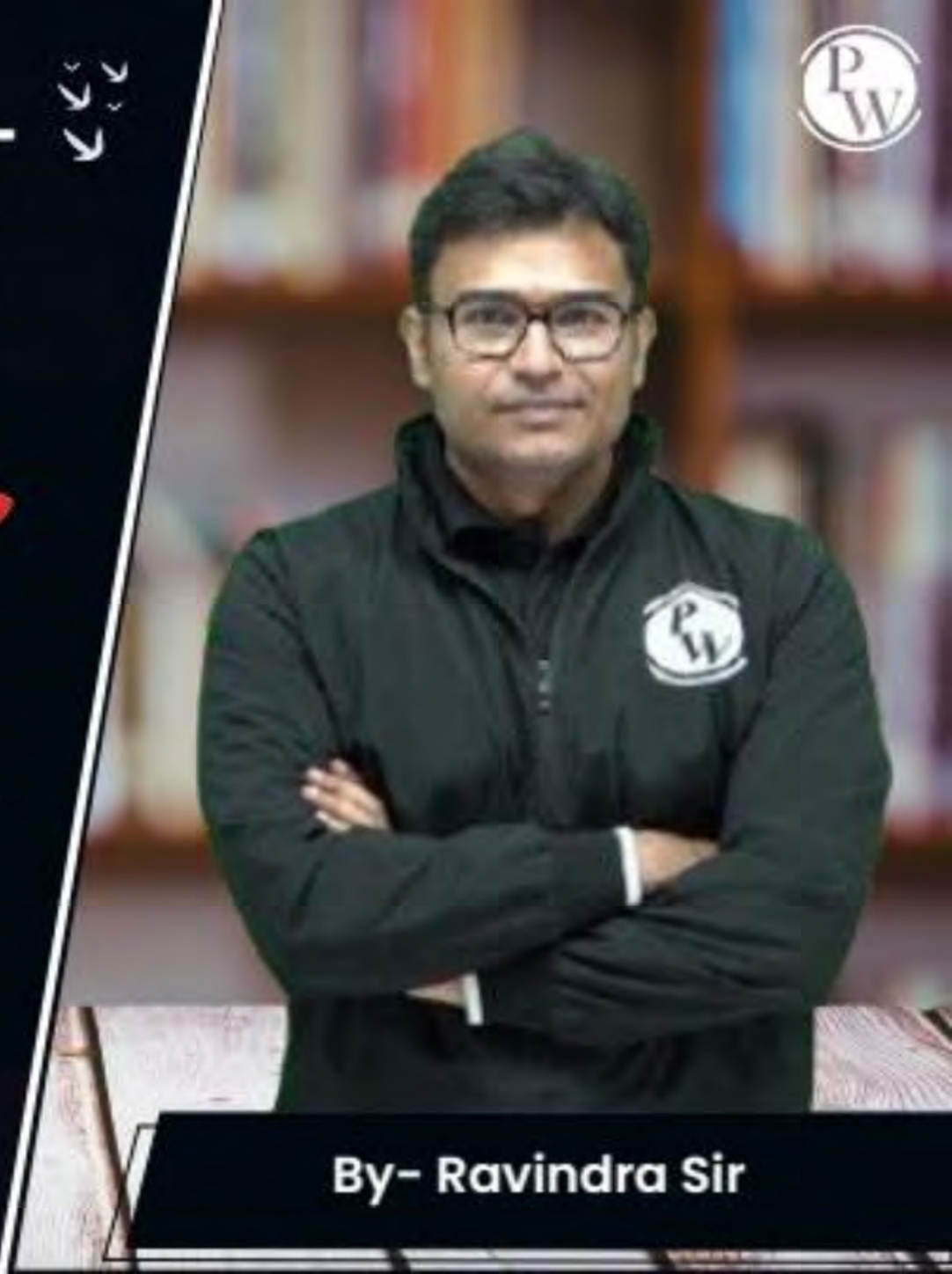
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 6



By- Ravindra Sir

Recap of Previous Lecture



- Topic — DFA, NFA, RE
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

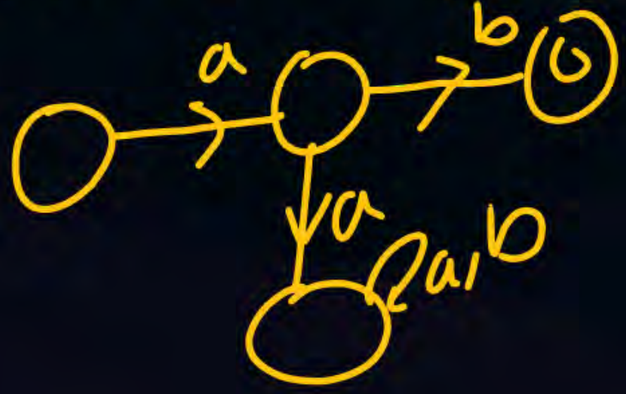
DFA, NFA & RE

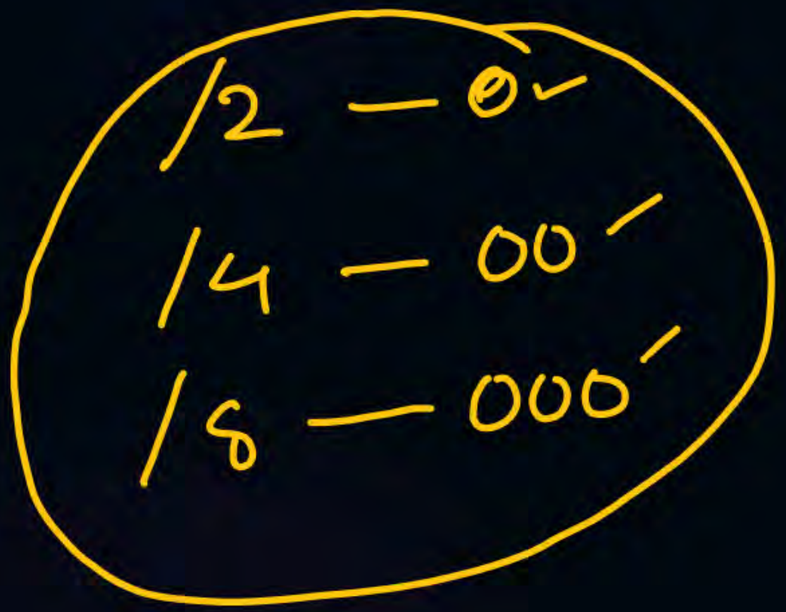
Properties of RE

Starting with → dead

Containing
Ending
> no dead

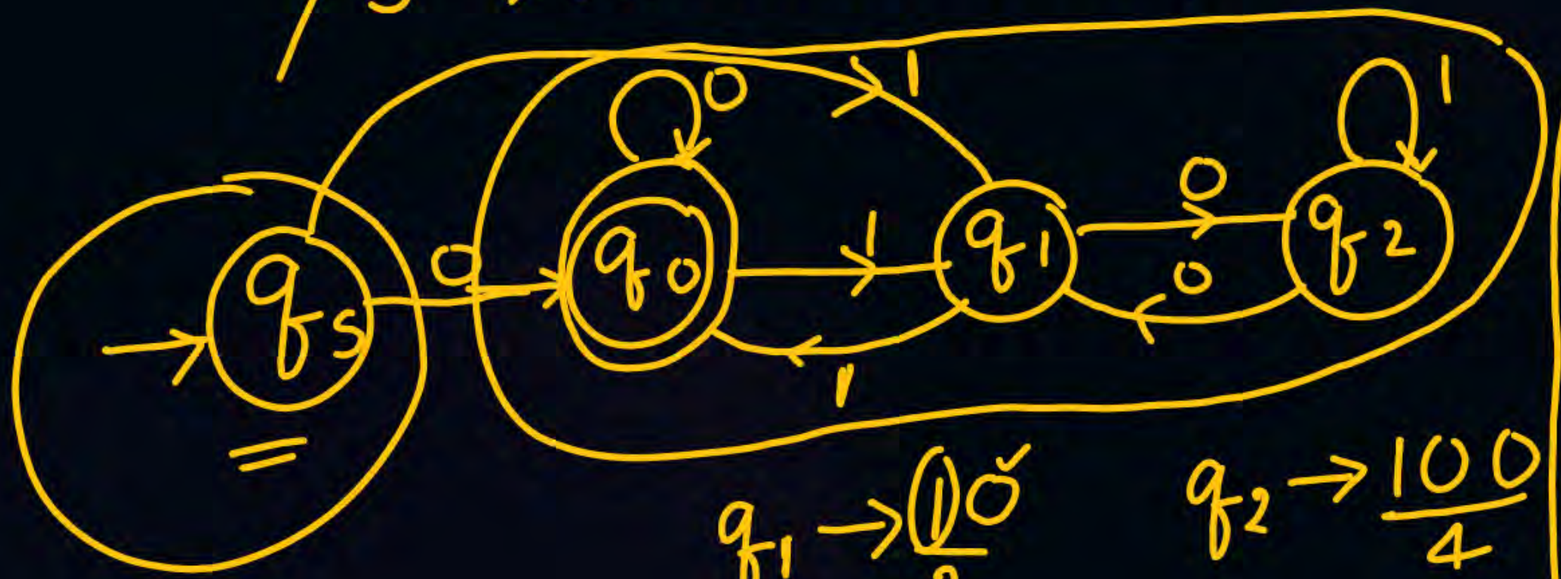
Every a should be followed by a 'b'




1/2 — 0'
1/4 — 00'
1/8 — 000'

$\Sigma = \{0, 1\}$ $L = \{\omega / \omega$ is a binary number divisible by $3\}$

/3 → remainders → 0, 1, 2



	0	1
*q_0	q_0	q_1
q_1	q_2	q_0
q_2	q_1	q_2
→q_s	q_0	q_1

$q_1 \rightarrow \frac{10}{2}$ $q_2 \rightarrow \frac{100}{4}$

$q_1 \rightarrow \frac{11}{3}$ $q_2 \rightarrow \frac{101}{5}$



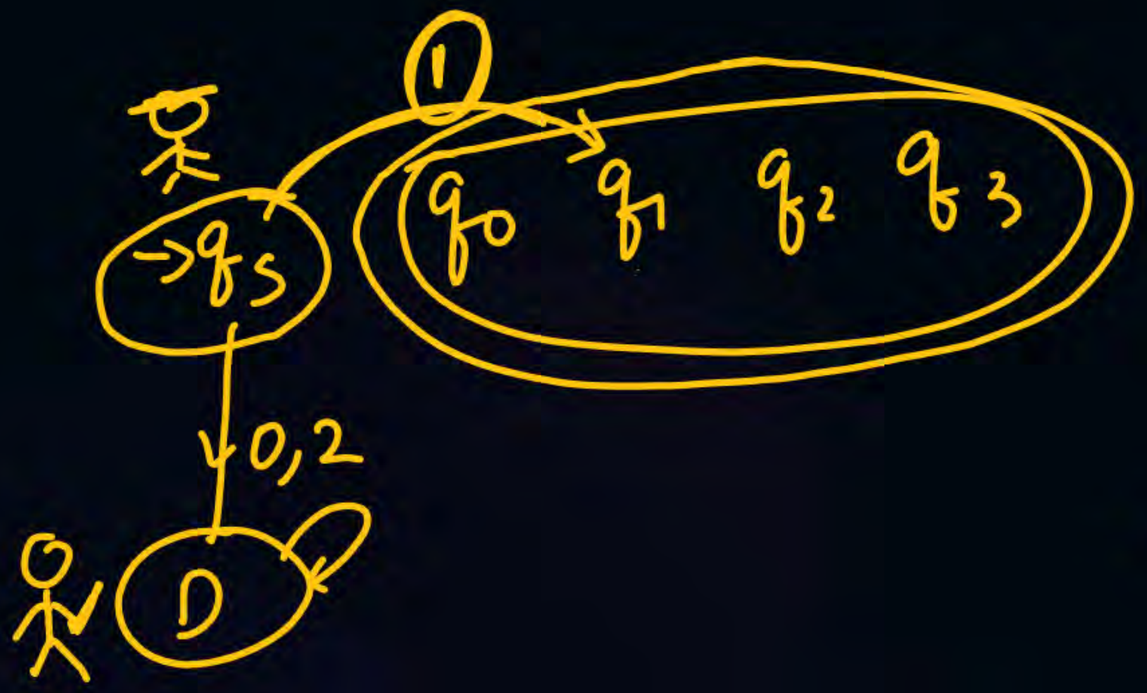
(ε)

$\Sigma = \{0, 1\}$ $L = \{ w / w$ is a binary number divisible by $5 \}$

q_0 q_1 q_2 q_3 q_4

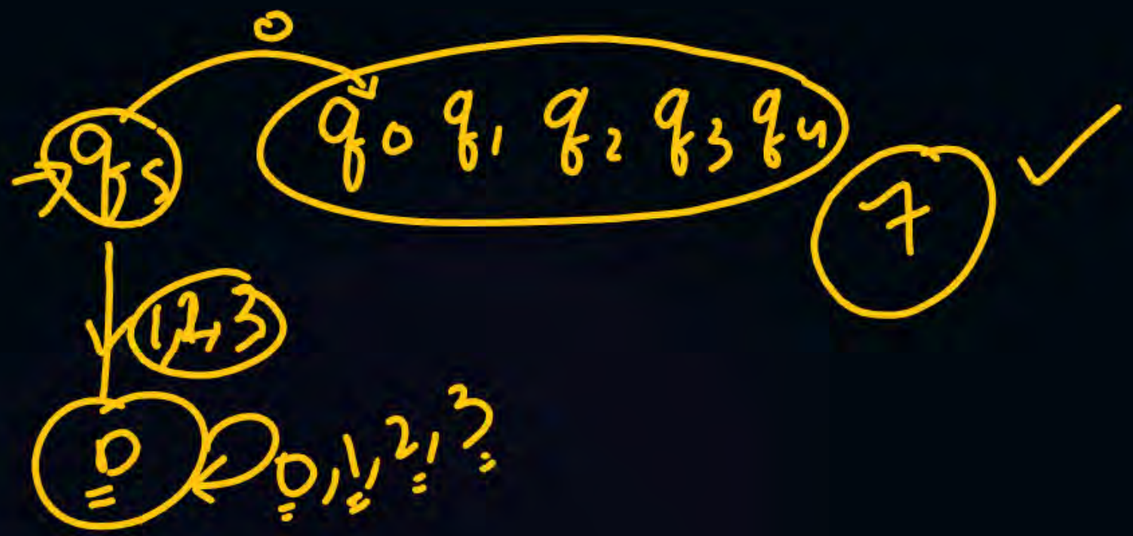
	0	1
$*q_0$	q_0	q_1
q_1	q_2	q_3
q_2	q_4	q_0
q_3	q_1	q_2
q_4	q_3	q_4
$\rightarrow \underline{q_5}$	q_0	q_1

$\Sigma = \{0,1,2\}$ $L = \{w / w$ is a base 3 number (divisible by 4) and starts with '1'$\}$



	0	1	2
$*q_0$	q_0	q_1	q_2
q_1	q_3	q_0	q_1
q_2	q_2	q_3	q_0
q_3	q_1	q_2	q_3
$\rightarrow q_s$	D	q_1	D
D	D	D	D

$\Sigma = \{0,1,2,3\}$ $L = \{\omega / \omega$ begins with 0 and is divisible by 5$\}$



	0	1	2	3
*q_0	q_0	q_1	q_2	q_3
q_1	q_4	q_0	q_1	q_2
q_2	q_3	q_4	q_0	q_1
q_3	q_2	q_3	q_4	q_0
q_4	q_1	q_2	q_3	q_4
→q_5	q_0	D	D	D
D	D	D	D	D

$\Sigma = \{0,1\}$ $L = \{ w / w$ is a binary number divisible by 4$\}$

00 ✓

2^k

3^k

4^k

3

5

6

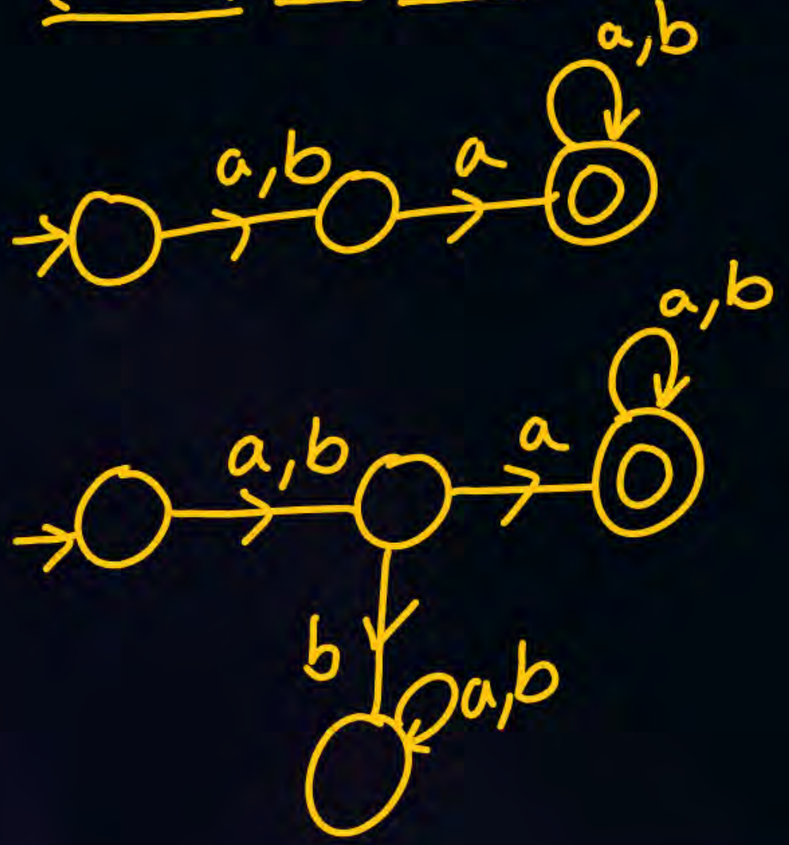
7

8X

=

$\Sigma = \{a, b\}$ $L = \{w / 2^{nd}$ Symbol in w is 'a' $\}$

$(a+b)(a)(a+b)^*$

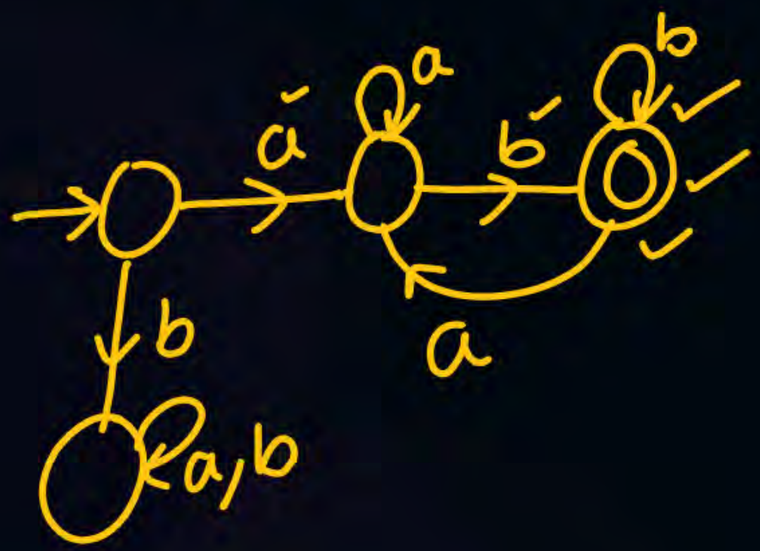


$\Sigma = \{a, b\}$ $L = \{ w / w$ starts with 'a' and ends with 'b' $\}$

$a(a+b)^*b$



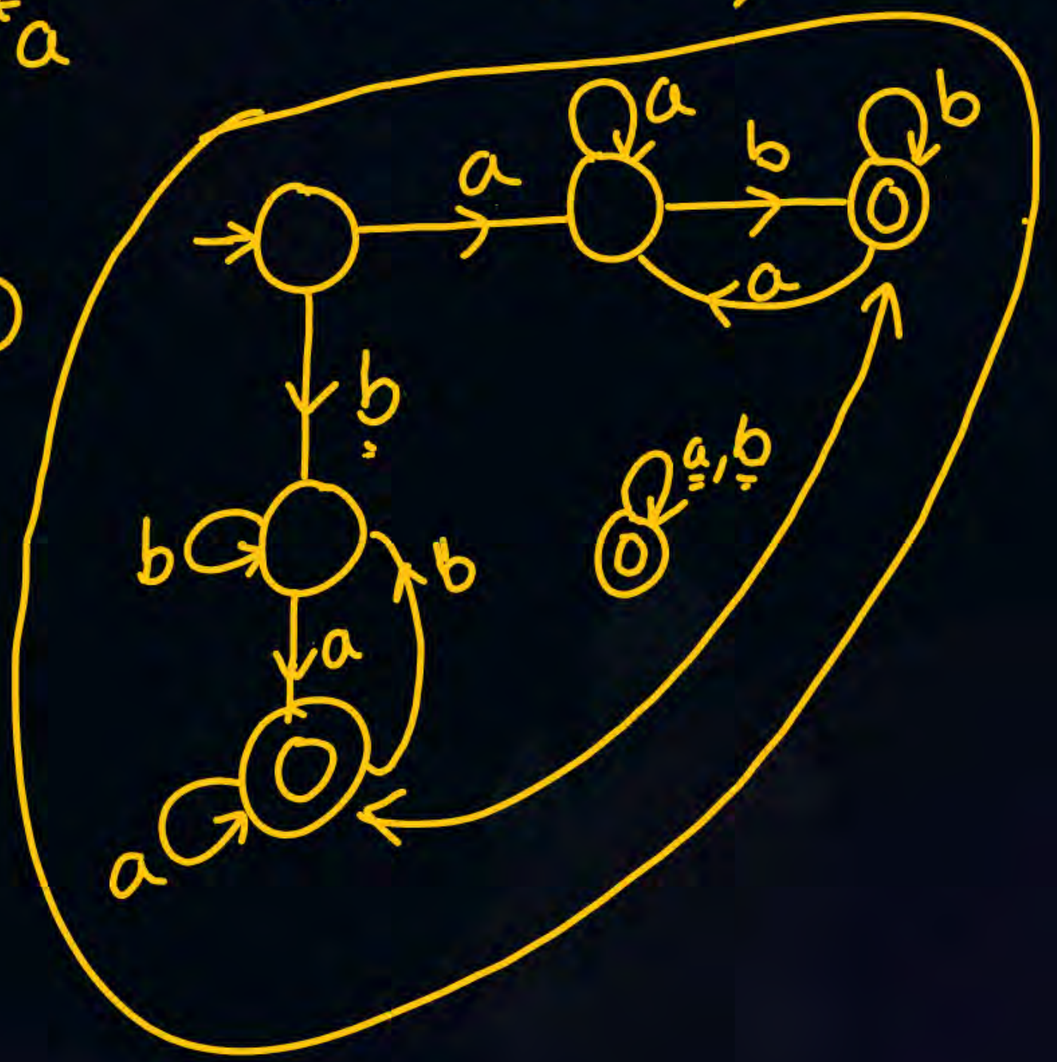
$L = \{ab, aab, abb \cdots\}$



$\Sigma = \{a, b\}$ $L = \{w / w$ start and ends with different symbol $\}$

$a(a+b)^*b + b(a+b)^*a$



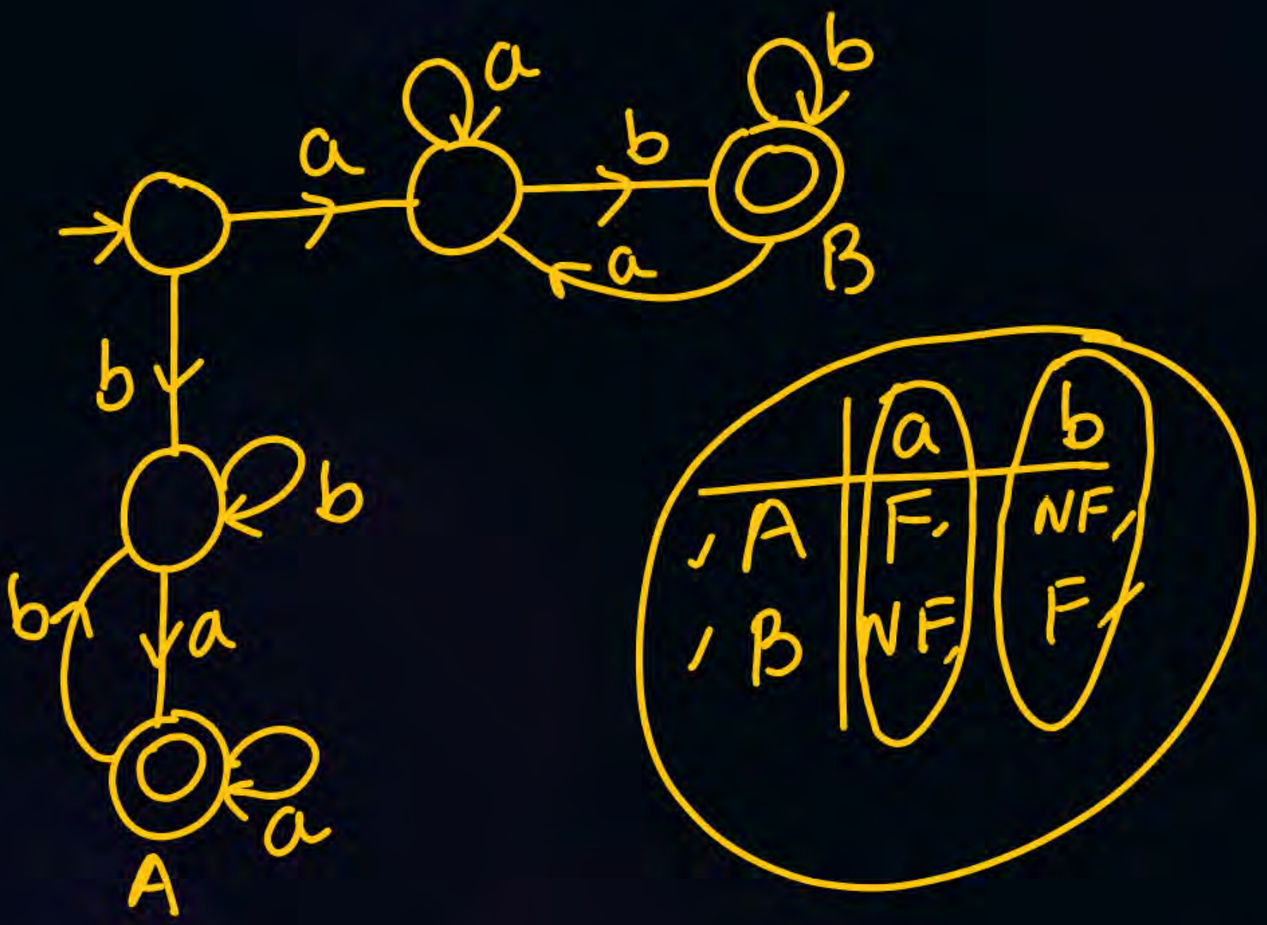


✓

min

DFA → min DFA

minimal

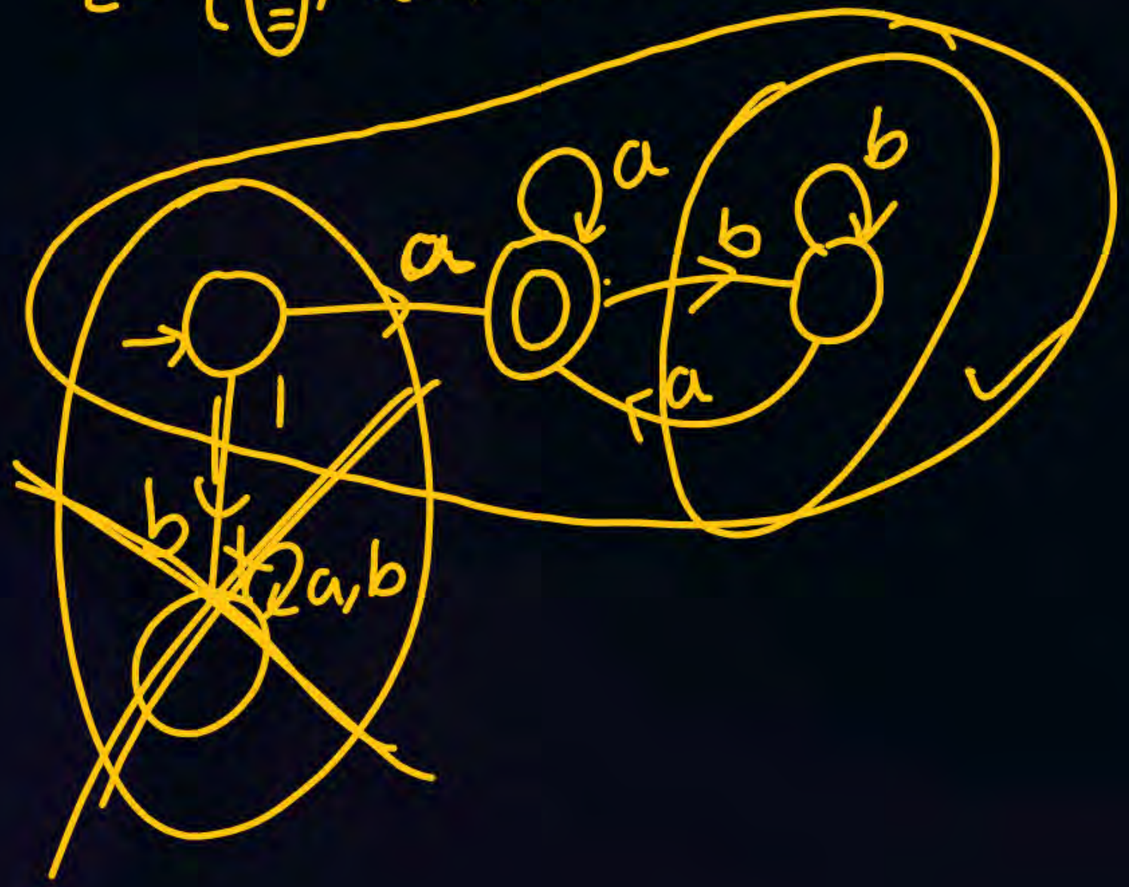


	a	b
A	F	NF
B	NF	F

$\Sigma = \{a, b\}$ $L = \{ w / w$ starts and ends with a $\}$

$a + a(a+b)^* a$

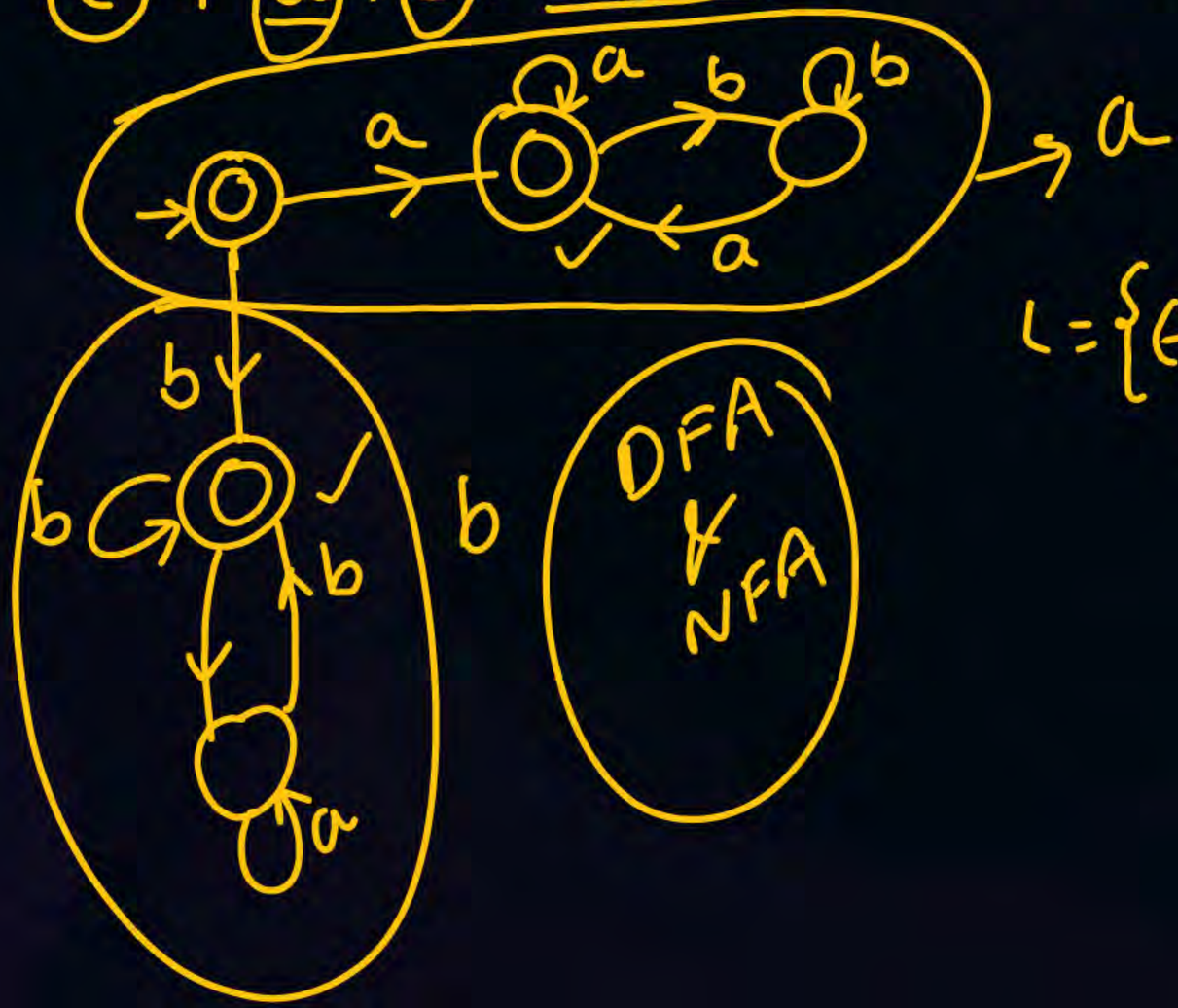
$L = \{a, aa, aaa, aba, \ldots\ldots \ldots\ldots\}$



starting and ending with same symbol

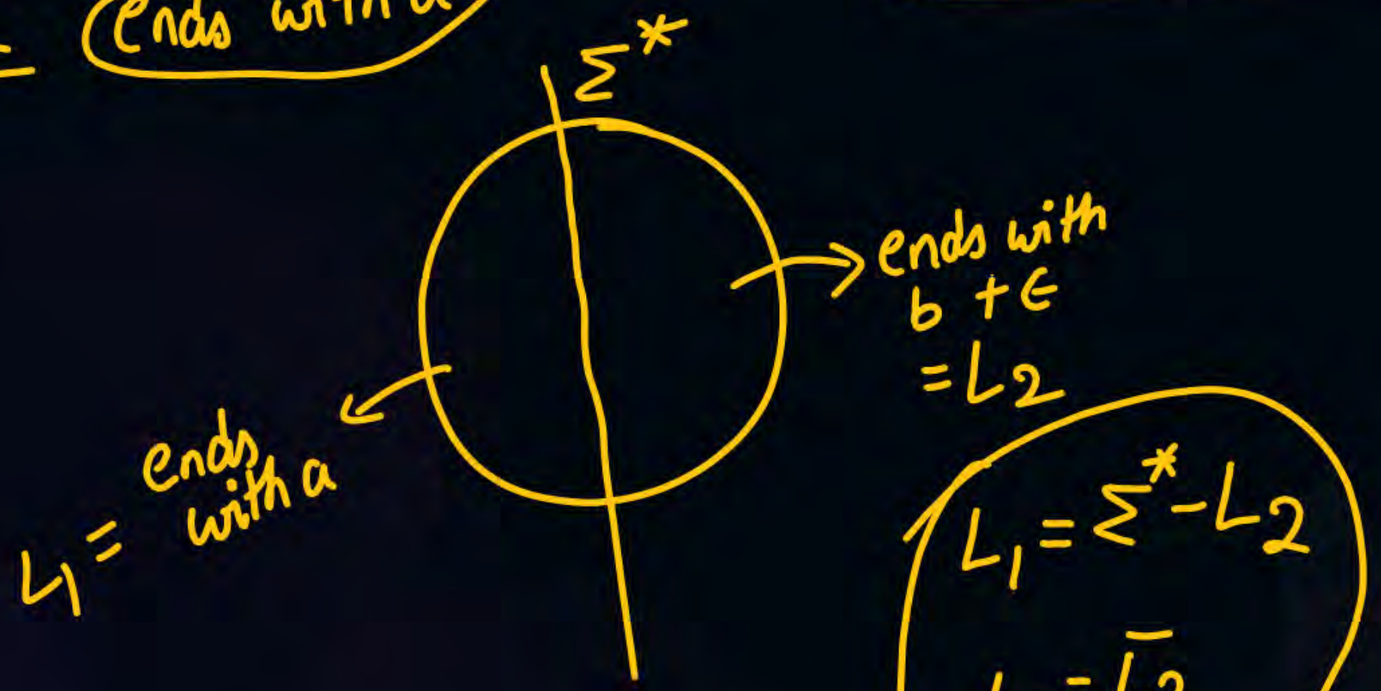
$\epsilon + a + b + a(a+b)^*a + b(a+b)^*b$

$\epsilon . \epsilon = \epsilon$



$L = \{\epsilon, a, b, aa, bb, aaa, aba, bab, \ldots\ldots\}$





$L_1 = \Sigma^* - L_2$

$L_1 = \bar{L_2}$

$L_2 = \bar{L_1}$

When we complement a DFA (make non final states as final and final states as non final), the language get complemented.

$(Q, \Sigma, \delta, q_0, F) \rightarrow$

$\Downarrow$ Comp

$(Q, \Sigma, \delta, q_0, Q-F) \rightarrow$



NFA but not DFA

ending with 'a'

Complement



$(a+b)^*$ → universal language

→ Complementing an NFA ~~does not~~ ✓ may / may not Complement the language.

THANK - YOU